Evaluating Prediction Intervals

Study Using the M3 Competition Dataset

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2024-11-27

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1 Avaliação da taxa de cobertura

1.1 Modelos sem *bagging*

1.1.1 STM

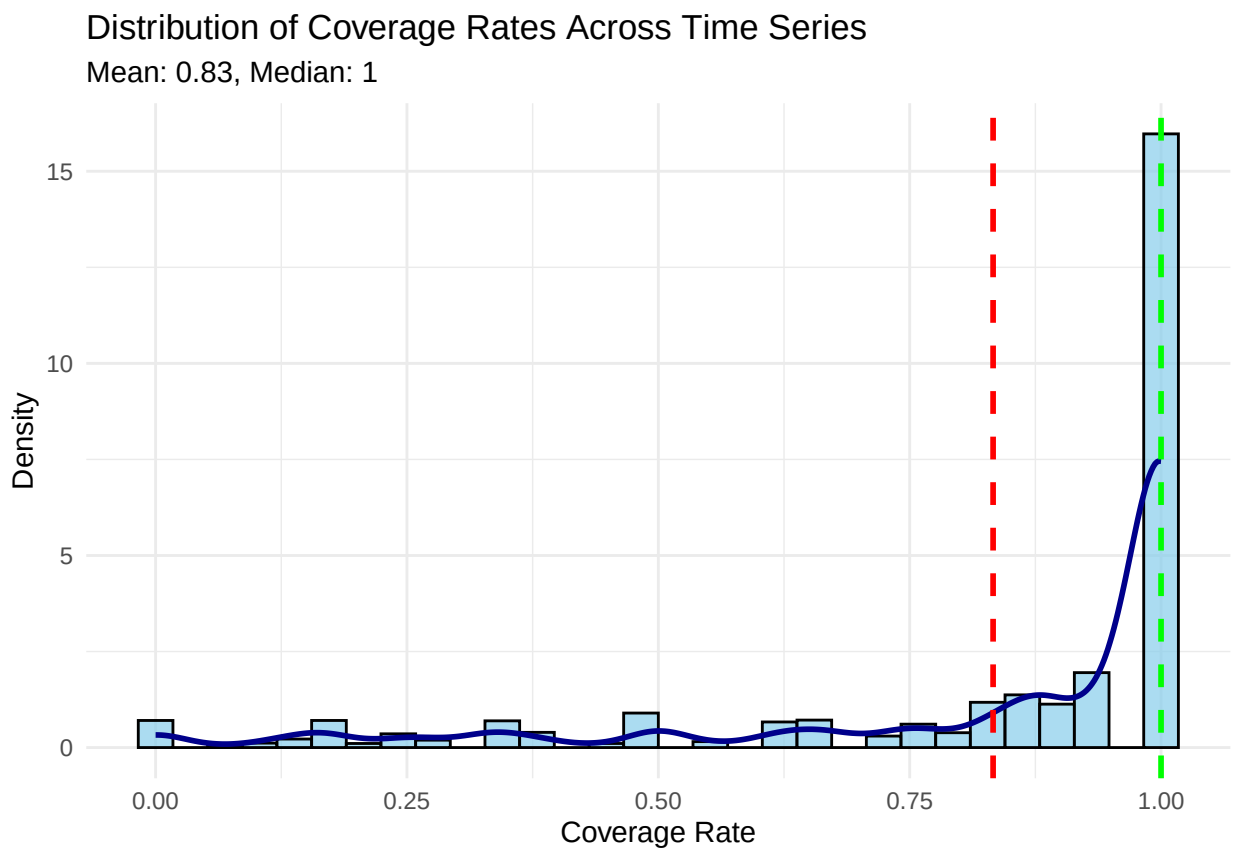
```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_stm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_stm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```

```
## Warning: The dot-dot notation (`..density..`) was deprecated in ggplot2 3.4.0.
## i Please use `after_stat(density)` instead.
## i The deprecated feature was likely used in the forecTheta package.
##   Please report the issue at <https://github.com/jafiorucci/forecTheta/issues>.
## This warning is displayed once every 8 hours.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```




```r
results$summary
```

```
## $mean
## [1] 0.8329726
##
## $median
## [1] 1
##
## $sd
## [1] 0.2724985
##
## $min
## [1] 0
##
## $max
## [1] 1
```

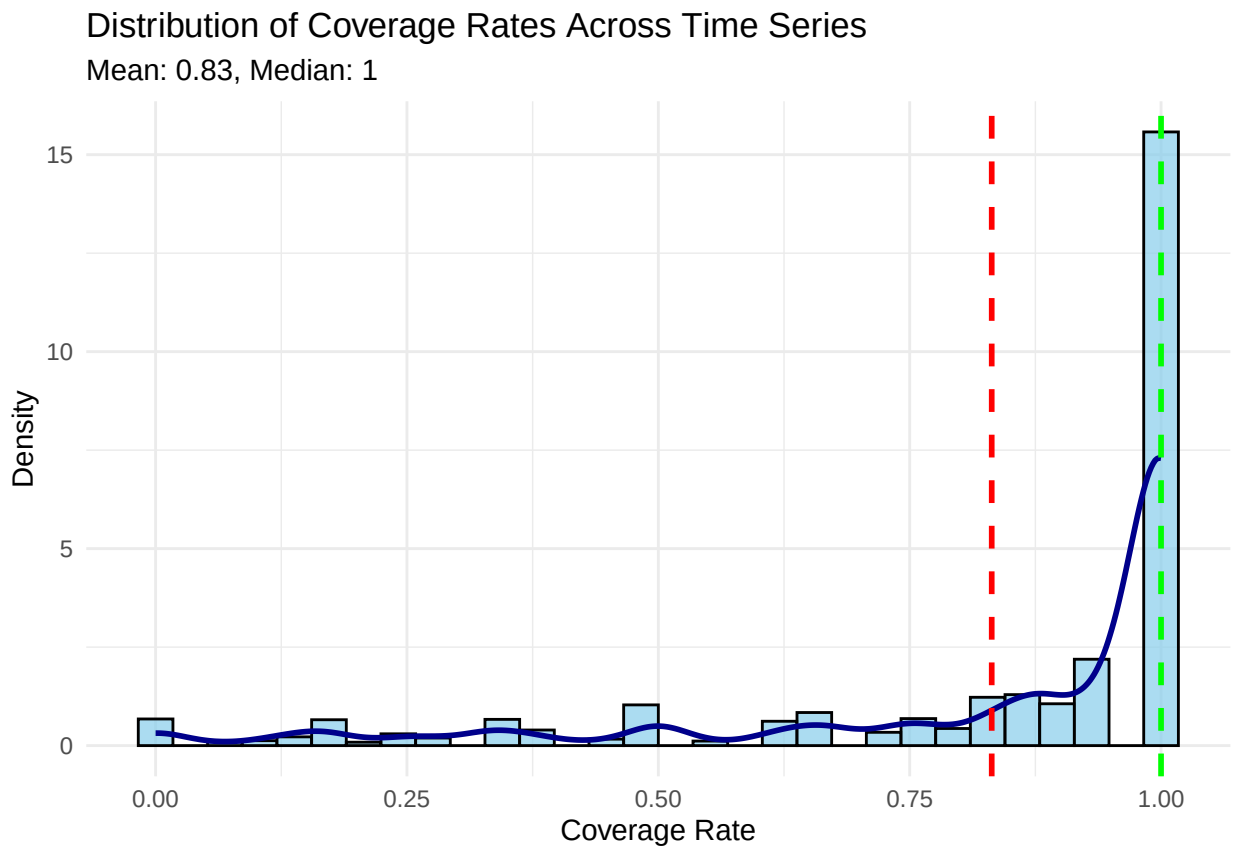
1.1.2 OTM

```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_otm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_otm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```
results$summary
```

```
## $mean
## [1] 0.8315943
##
```

```
## $median
## [1] 1
##
## $sd
## [1] 0.2700621
##
## $min
## [1] 0
##
## $max
## [1] 1
```

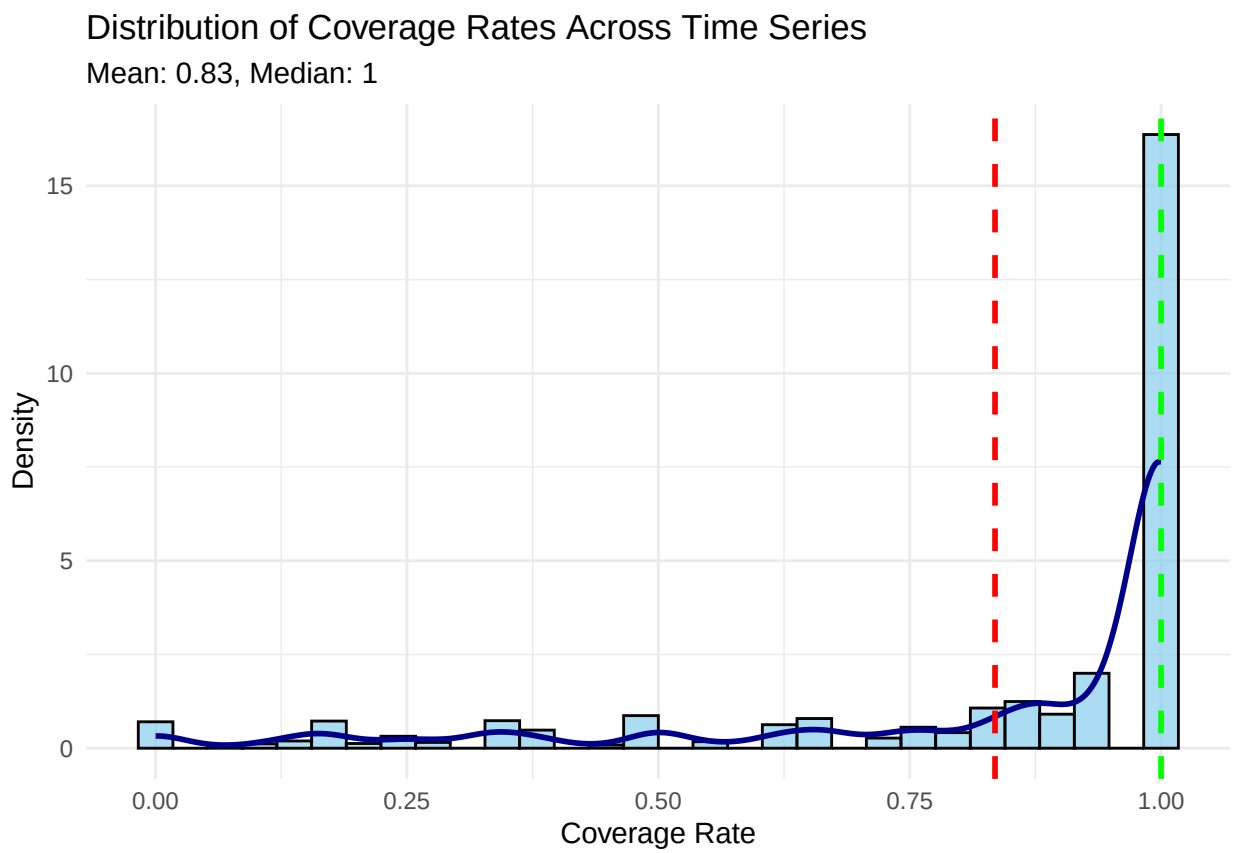
1.1.3 DSTM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_dstm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_dstm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```
results$summary
```

```
## $mean
## [1] 0.8348735
##
## $median
## [1] 1
##
## $sd
## [1] 0.2727318
##
## $min
## [1] 0
##
## $max
## [1] 1
```

1.1.4 DOTM

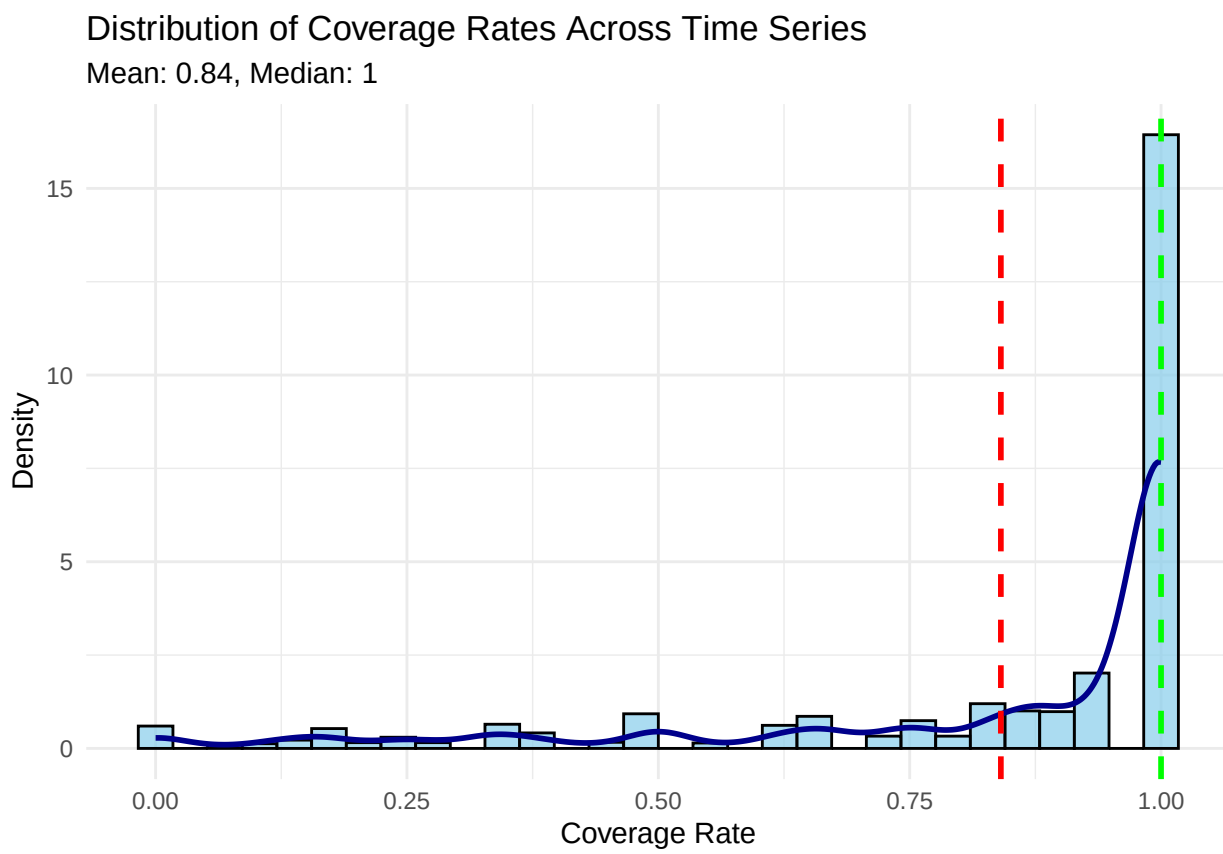
```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
```

```r
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_dotm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_dotm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```r
results$summary
```

```
## $mean
## [1] 0.8406825
##
## $median
## [1] 1
##
## $sd
## [1] 0.2644391
##
## $min
## [1] 0
```

```
##
## $max
## [1] 1
```

1.2 Modelos com *bagging*

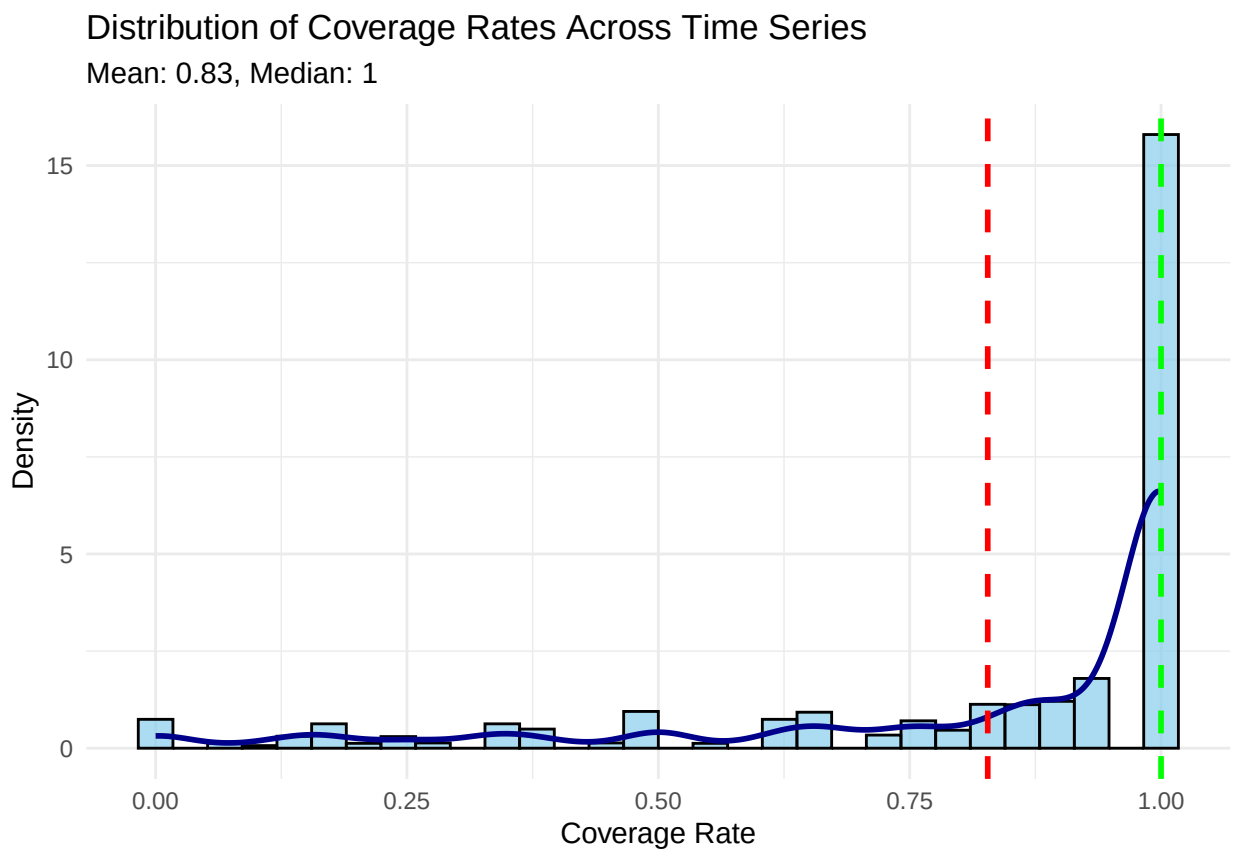
1.2.1 *Bagged* STM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_bagged_stm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_bagged_stm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```
results$summary
```

```
## $mean
## [1] 0.8276353
##
## $median
## [1] 1
##
## $sd
## [1] 0.2746695
##
## $min
## [1] 0
##
## $max
## [1] 1
```

1.2.2 *Bagged* OTM

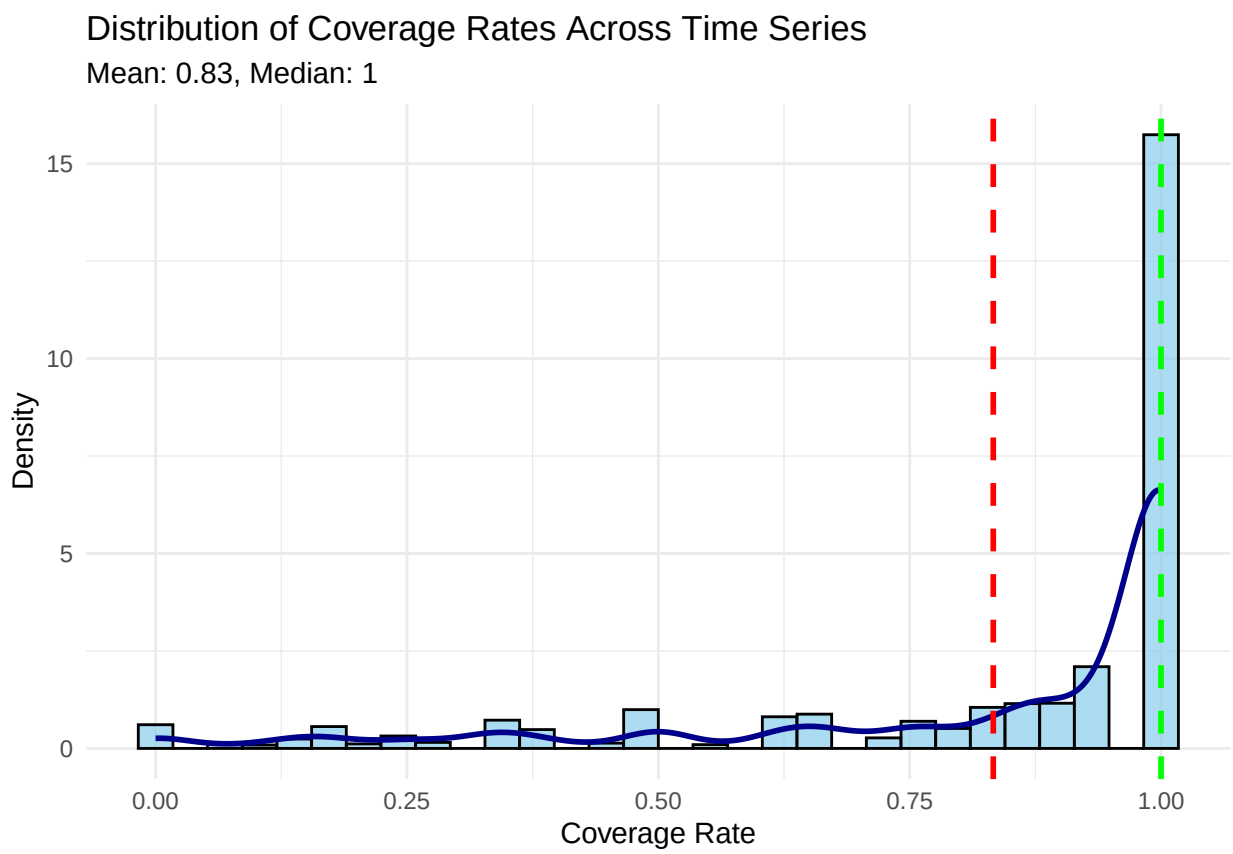
```
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
```

```r
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_bagged_otm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_bagged_otm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```r
results$summary
```

```
## $mean
## [1] 0.8331437
##
## $median
## [1] 1
##
## $sd
## [1] 0.266965
##
## $min
## [1] 0
```

```
##
## $max
## [1] 1
```

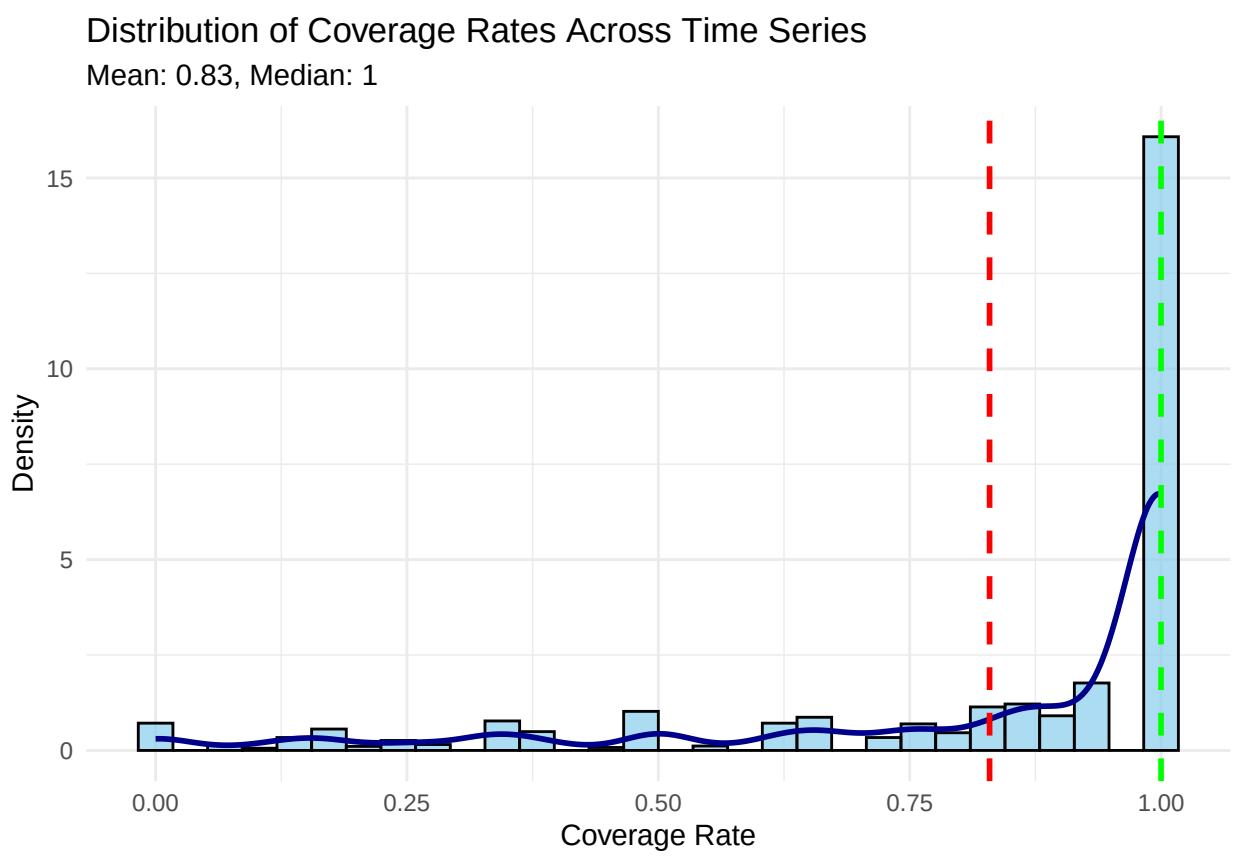
1.2.3 *Bagged* DSTM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_bagged_dstm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_bagged_dstm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```



```r
results$summary
```

```
## $mean
## [1] 0.8294622
##
## $median
## [1] 1
##
## $sd
## [1] 0.2737066
##
## $min
## [1] 0
##
## $max
## [1] 1
```

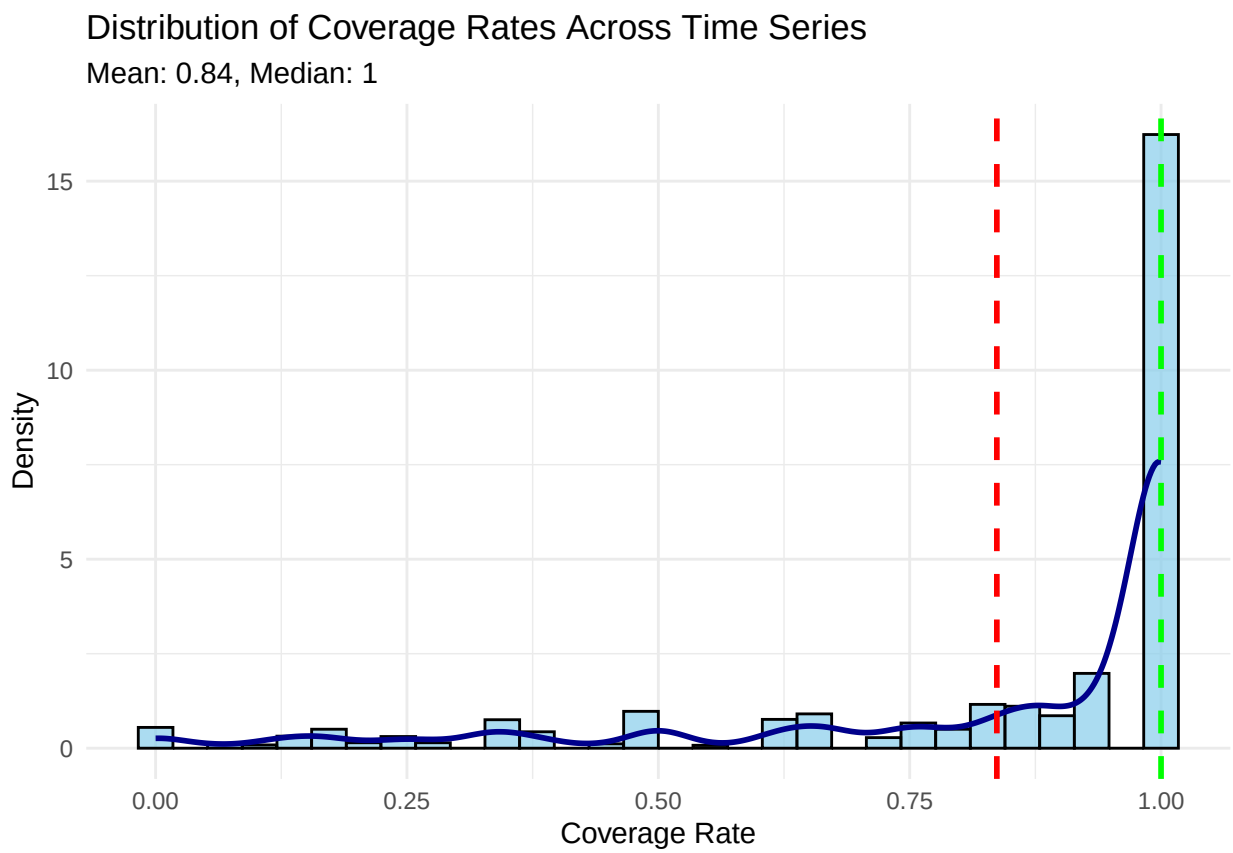
1.2.4 *Bagged* DOTM

```r
lo <- matrix(NA, nrow=3003, ncol=18)
hi <- matrix(NA, nrow=3003, ncol=18)
obs = matrix(NA, nrow=3003, ncol=18) #matrix of the out-sample values

coverage_rates <- c()
for(i in 1:3003){
  x=M3[[i]]$x
  h=M3[[i]]$h
  obs[i, 1:h] = M3[[i]]$xx
  lo[i, 1:h] = out_bagged_dotm[[i]]$lower[, 3] #.95 interval
  hi[i, 1:h] = out_bagged_dotm[[i]]$upper[, 3] #.95 interval

  coverage_rates <- c(
    coverage_rates,
    calculate_coverage_rate(obs[i, 1:h], lo[i, 1:h], hi[i, 1:h])
  )
}

results <- evaluate_coverage_rates(coverage_rates)
```




```r
results$summary
```

```
## $mean
## [1] 0.8366865
##
## $median
## [1] 1
##
## $sd
## [1] 0.2658834
##
## $min
## [1] 0
##
## $max
## [1] 1
```